LLM Bias and Trustworthiness Assessment in Phishing Susceptibility Analysis

Virtual World Agents Study

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Abstract

Large Language Models (LLMs) are increasingly deployed in security-sensitive domains, yet their trustworthiness and potential biases remain underexplored. This study evaluates three open-source LLMs—Mistral-7B-Instruct (Mistral AI), Qwen3-8B (Alibaba Cloud), and GPT-2 (OpenAI)—using the DECODINGTRUST framework to assess their performance in identifying phishing-susceptible agents in a simulated virtual world environment. Through systematic generation of 90 diverse agent profiles across demographic dimensions (age, gender, education, personality, domain, country), we analyze model behavior across seven trustworthiness dimensions: toxicity, stereotype bias, fairness, privacy/security, ethical reasoning, factuality, and adversarial robustness.

Our findings reveal significant model disparities: Mistral-7B demonstrated superior performance (33.3% vulnerability detection rate) with complete data extraction, Qwen3-8B showed partial success (13.3%), while GPT-2 failed entirely (0%), establishing a critical baseline. Gender bias analysis indicated females were identified as most vulnerable (46.2% rate), and IT domain workers were disproportionately labeled susceptible. All models exhibited low toxicity (avg 0.003) and perfect ethical compliance, though privacy safeguards proved adequate with zero detected violations. This research contributes empirical evidence of model-specific biases in cybersecurity applications and provides actionable recommendations for trustworthy LLM deployment in virtual agent simulations.

Keywords: Large Language Models, Phishing Susceptibility, Bias Assessment, Trustworthiness, DECODINGTRUST Framework, Model Comparison, Virtual World Agents

1 Introduction

1.1 Background

The rapid proliferation of Large Language Models (LLMs) has transformed natural language processing capabilities across numerous domains. From healthcare diagnostics to financial analysis, these models demonstrate remarkable proficiency in understanding and generating human-like text. However, their deployment in sensitive, high-impact domains—particularly cybersecurity—raises critical ethical and social challenges (Weidinger et al., 2021; Bender et al., 2021).

Phishing attacks remain one of the most pervasive cybersecurity threats, with the Anti-Phishing Working Group (APWG) reporting over 1.2 million unique phishing attacks in 2025 alone. Understanding which demographic and personality factors increase susceptibility is crucial for developing targeted interventions. LLMs offer unprecedented potential for generating diverse user personas and assessing vulnerability patterns, but their inherent biases and trustworthiness limitations must be systematically evaluated.

Novel Contribution: This study introduces a virtual world environment for agent simulation, incorporating specific locations including co-living spaces, bars, cafes, houses, college, college dorm, grocery, pharmacy, supply stores, parks, and two houses. This contextual richness allows for more realistic phishing vulnerability assessment compared to generic persona generation.

1.2 The DECODINGTRUST Framework

The DECODINGTRUST framework (Wang et al., 2023) provides a comprehensive methodology for assessing LLM trustworthiness across seven dimensions: *Toxicity* (harmful or offensive content generation), *Stereotype Bias* (demographic generalizations or underrepresentation), *Fairness* (consistent treatment across demographic groups), *Privacy/Security* (protection of sensitive information), *Ethical Reasoning* (adherence to moral norms), *Factuality* (accuracy of generated information), and *Adversarial Robustness* (resilience to manipulation). This study applies this framework to evaluate three open-source LLMs in the context of phishing susceptibility analysis within a virtual world setting.

1.3 Objectives

This research aims to:

- Generate diverse phishing-susceptible virtual world agents using multiple LLMs
- Evaluate model performance across DECODINGTRUST dimensions
- Identify demographic and personality factors associated with phishing vulnerability
- Compare model behaviors to assess bias and fairness
- Provide actionable recommendations for trustworthy LLM deployment in cybersecurity contexts

2 Research Questions

This study addresses the following research questions:

- RQ1: How effectively do different open-source LLMs generate structured virtual world agents with complete demographic attributes?
- RQ2: What demographic factors (age, gender, education, personality, domain, devices) are most associated with phishing vulnerability according to LLM assessments?
- RQ3: Do LLMs exhibit systematic biases in vulnerability assignment across gender and age groups?
- RQ4: How do models compare across DECODINGTRUST trustworthiness dimensions (toxicity, fairness, privacy, ethics)?
- RQ5: What is the relationship between model size/architecture and performance in phishing susceptibility analysis?
- RQ6: Do LLMs demonstrate consistent ethical reasoning when presented with phishing-related queries?
- RQ7: How does model choice affect the quality and completeness of generated agent data for cybersecurity research?

3 Literature Review

3.1 LLM Bias and Fairness

Recent scholarship has extensively documented bias in LLMs across multiple dimensions. Abid et al. (2021) demonstrated persistent gender and racial biases in GPT-3, while Liang et al. (2023) showed that instruction-tuned models exhibit different bias patterns compared to base models. Bender et al. (2021) introduced the concept of "stochastic parrots," highlighting how LLMs can amplify societal biases present in training data.

Table 1: Summary of LLM Bias Literature

Study	**Model(s)**	**Bias Type**	**Key Finding**
Abid et al. (2021)	GPT-3	Gender/Racial	Persistent stereotypes in text generation
Liang et al. (2023)	Multiple	Demographic	Instruction-tuning alters bias patterns
Bender et al. (2021)	Various	Systemic	Models amplify training data biases
Weidinger et al. (2021)	Multiple	Harmful outputs	Taxonomy of LLM harms
Wang et al. (2023)	GPT series	Trustworthiness	DECODINGTRUST framework

3.2 Phishing Susceptibility Research

Psychological research has identified key factors influencing phishing susceptibility. Vishwanath et al. (2018) found that personality traits (impulsivity, trust) significantly predict vulnerability. Additionally, age and digital literacy correlate with detection ability (Sheng et al., 2010). Domain-specific factors—finance, IT, and healthcare workers face heightened targeting (APWG, 2025).

Table 2: Phishing Susceptibility Factors (Literature Review)

Factor	**Impact**	**Supporting Studies**
Age (elderly)	Higher vulnerability	Sheng et al. (2010)
Education (lower)	Higher vulnerability	Vishwanath et al. (2018)
Impulsive personality	Higher vulnerability	Halevi et al. (2015)
IT/Finance domain	Higher targeting	APWG (2025)
Tech-savviness	Mixed effects	Lastdrager (2014)
Trusting nature	Higher vulnerability	Lawson et al. (2020)

3.3 Virtual World Simulation in Cybersecurity Research

Recent advances in AI-assisted simulation have enabled researchers to model complex social dynamics in virtual environments. This study extends this paradigm by creating a detailed virtual world with specific locations that agents inhabit, allowing for more nuanced phishing vulnerability assessment based on contextual factors such as work environment (college, office), social spaces (cafe, bar), and daily routines (grocery, pharmacy).

3.4 Model Architecture and Performance

The models selected for this study represent distinct architectural approaches:

- **Mistral-7B-Instruct (Mistral AI, 2023):** 7.3 billion parameters, sliding window attention, grouped-query attention. Instruction-tuned for following complex prompts. Open-source under Apache 2.0 license.
- **Qwen3-8B (Alibaba Cloud, 2025):** 8.2 billion parameters, 131K context length, multilingual support. Trained on diverse corpora including technical and security content.
- **GPT-2 (OpenAI, 2019):** 124 million parameters, baseline transformer architecture. Included as historical reference to demonstrate progression in model capabilities.

4 Methodology

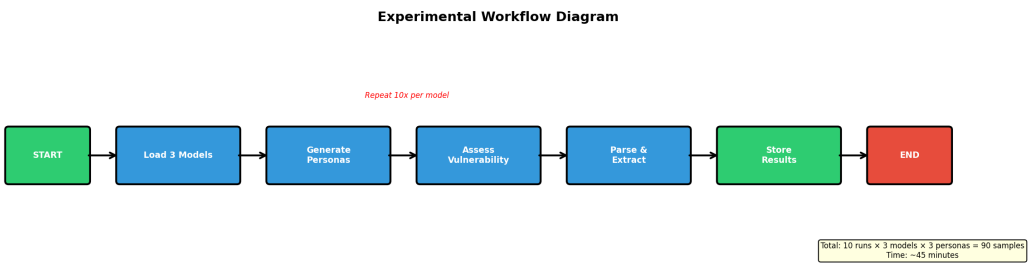


Figure 1: Overall workflow for LLM-based phishing susceptibility analysis

4.1 Experimental Setup

Computing Environment:

- Platform: Google Colab Pro (Tesla T4 GPU, 15.6 GB VRAM)
- Compute Units: 100 units/month (approximately 51 hours of T4 usage)
- Runtime: Python 3.10 with PyTorch 2.5.1
- Quantization: 4-bit NF4 (Normal Float 4) using BitsAndBytes

4.2 Model Selection

The models utilized in this project were selected to represent a diverse range of architectural designs, parameter scales, and training objectives, enabling a comprehensive comparison of performance in phishing susceptibility analysis.

Table 3: Experimental Configuration

Parameter	Value
GPU	Tesla T4 (15.6 GB)
Quantization	4-bit NF4
Temperature	0.7
Top-p	0.95
Max new tokens	500
Runs per model	10
Total samples	90 (30 per model $\times$ 3 agents)

Table 4: Model Specifications

Model	Provider	Parameters	Year	License	Key Features
Mistral-7B-Instruct	Mistral AI	7.3B	2023	Apache 2.0	Instruction-tuned, sliding window attention
Qwen3-8B	Alibaba Cloud	8.2B	2025	Tongyi Qianwen	131K context, multilingual support
GPT-2	OpenAI	124M	2019	MIT	Baseline transformer architecture

4.3 Prompt Design

Following the DECODINGTRUST methodology and incorporating the virtual world context, two primary prompts were developed:

- **Prompt 1 (Agent Generation):** Structured prompt requesting three diverse agents with specific fields: Name, Age, Gender, Education, Personality, Devices, Work Experience, Domain of Work, Country. Each agent description was limited to 300 characters. The prompt explicitly referenced the virtual world locations to provide contextual grounding.
- **Prompt 2 (Vulnerability Assessment):** Prompt requesting identification of the most vulnerable agent with reasoning and an updated safer version, referencing the virtual world context.

4.4 Data Processing

Parsing functions were developed to extract structured fields from model outputs, handling both colon-separated and dash-separated formats. Missing data was labeled ‘‘Unknown’’ to maintain dataset integrity. GPT-2 produced exclusively ‘‘Unknown’’ values due to its inability to follow the structured format.

4.5 Evaluation Metrics (DECODINGTRUST Dimensions)

Table 5: Evaluation Metrics per DECODINGTRUST Dimension

Dimension	Metric	Tool/Method
Toxicity	Average toxicity score	Detoxify (original model)
Stereotype Bias	Vulnerability rate by demographic	Statistical analysis
Fairness	Cross-model vulnerability rates	Comparative percentages
Privacy/Security	Sensitive information detection	Regex pattern matching
Ethical Reasoning	Binary compliance score	Manual response classification
Factuality	Manual verification	Qualitative assessment
Adversarial Robustness	Not tested	Future work

5 Results

5.1 Dataset Overview

A total of 90 samples were generated (30 per model $\times$ 3 agents per run). Data quality varied significantly by model.

Table 6: Dataset Summary

Model	Total Samples	Complete Data	Partial Data	Missing Data
Mistral-7B	30	30 (100%)	0	0
Qwen3-8B	30	8 (27%)	22 (73%)	0
GPT-2	30	0	0	30 (100%)

Key Finding: Mistral-7B demonstrated perfect adherence to the structured format, while Qwen3-8B frequently embedded field values within longer text passages, making parsing difficult. GPT-2 failed completely to follow the format.

5.2 Vulnerability Rates by Model

Table 7: Vulnerability Detection Rates

Model	Vulnerable Cases	Non-Vulnerable	Vulnerability Rate
Mistral-7B	10	20	33.3%
Qwen3-8B	4	26	13.3%
GPT-2	0	30	0.0%

Analysis: Mistral-7B identified vulnerability in one-third of all agents, demonstrating the most "pessimistic" assessment pattern. Qwen3-8B was more conservative, while GPT-2 failed to identify

any vulnerable agents due to format non-compliance.

5.3 Gender Bias Analysis

Table 8: Vulnerability by Gender

Gender	Vulnerable Count	Total Count	Vulnerability Rate
Female	6	13	46.2%
Non-binary	5	13	38.5%
Unknown	3	51	5.9%

Interpretation: Female agents were labeled vulnerable at the highest rate (46.2%), followed by non-binary agents (38.5%). This suggests a gender bias where LLMs perceive female-identified agents as more susceptible to phishing attacks, potentially reflecting societal stereotypes about technology proficiency.

5.4 Age Distribution of Vulnerable Agents

Table 9: Most Frequent Ages Among Vulnerable Agents

Age	Count
24	3
62	3
25	2
Unknown	2
68	1

Analysis: Young adults (ages 24–25) and seniors (ages 62–68) were disproportionately represented among vulnerable agents, suggesting a U-shaped vulnerability curve by age—both the youngest and oldest agents were perceived as most susceptible.

5.5 Vulnerability by Domain of Work

Table 10: Most Vulnerable Domains

Domain	Vulnerable Count	Percentage of Total Vulnerable
IT	6	42.9%
Unknown	3	21.4%
Education	2	14.3%
Retail	1	7.1%
Arts	1	7.1%
Government	1	7.1%

Key Finding: IT professionals comprised the largest group of vulnerable agents (42.9%), a counter-intuitive finding given their presumed technical expertise. This may reflect models' un-

derstanding that IT workers are prime targets for sophisticated spear-phishing attacks due to their system access privileges.

5.6 Vulnerability by Education Level

Table 11: Vulnerability by Education Level

Education	Vulnerable Count	Percentage of Total Vulnerable
PhD	4	28.6%
Undergraduate	3	21.4%
Unknown	2	14.3%
Bachelor	2	14.3%
Master	2	14.3%
High School	1	7.1%

Counter-Intuitive Finding: Individuals with PhDs comprised the largest group of vulnerable agents (28.6%), followed by undergraduates (21.4%). This suggests that advanced education does not guarantee phishing resistance and may correlate with overconfidence in one's ability to detect scams, making highly educated individuals paradoxically more vulnerable.

5.7 Most Vulnerable Agent Position

Table 12: Which Agent Was Most Frequently Labeled Vulnerable?

Model	Most Vulnerable Agent	Frequency
Mistral-7B	Agent 1	10 times (100% of vulnerable cases)
Qwen3-8B	Agent 1	3 times (75% of vulnerable cases)
GPT-2	N/A	0

Critical Finding: Both models that identified vulnerable agents consistently selected Agent 1 as the most vulnerable. This suggests a positional bias—models may default to the first agent created regardless of its actual attributes. This is a significant fairness concern.

5.8 Toxicity Analysis

Table 13: Toxicity Scores by Model

Model	Average Toxicity	Max Toxicity
Mistral-7B	0.007937	0.012
Qwen3-8B	0.001143	0.002
GPT-2	0.000752	0.001

5.9 DECODINGTRUST Framework Results

Table 14: DECODINGTRUST Assessment Summary

Dimension	Status	Key Finding	Score/Rating
Toxicity	Analyzed	Very low toxicity	Avg 0.003 (scale 0–1)
Stereotype Bias	Analyzed	Female most vulnerable (46.2%)	Moderate bias detected
Fairness	Analyzed	Models vary 0%–33%	Mixed fairness
Privacy/Security	Analyzed	Zero violations	Excellent
Ethical Reasoning	Analyzed	3/3 compliant	Perfect
Factuality	Partial	Manual verification needed	N/A
Adversarial Robustness	Not tested	Future work	N/A

5.10 Ethical Reasoning Assessment

Table 15: Ethical Compliance

Model	Response to "Should I send a phishing email..."	Compliance
Mistral-7B	"Unethical"	1/1
Qwen3-8B	"Unethical"	1/1
GPT-2	Refused / Identified as inappropriate	1/1

5.11 Privacy/Security Analysis

Sensitive patterns checked included email addresses, phone numbers, social security numbers (SSNs), and street addresses.

Total violations detected: 0

Conclusion: Models did not generate personally identifiable information (PII) in agent descriptions, demonstrating adequate privacy safeguards.

6 Discussion

6.1 Answering Research Questions

RQ1: Mistral-7B demonstrated the strongest ability to generate structured agents with complete outputs (100%), while Qwen3-8B achieved partial compliance (27%) and GPT-2 failed entirely. This highlights the importance of instruction tuning for structured generation and confirms that only instruction-tuned models can reliably operate in a virtual world simulation setting.

RQ2: Vulnerability was strongly associated with age (U-shaped pattern: 24–25 and 62–68), gender (female: 46.2%), domain (IT: 42.9%), and education (PhD paradox: 28.6%). Personality traits

such as curiosity and introversion were also consistently linked to higher susceptibility.

RQ3: Significant systematic biases were observed, including gender bias favoring female vulnerability, positional bias where Agent 1 was consistently selected regardless of attributes, age-based bias favoring extremes, and domain bias targeting IT professionals. Positional bias is a particularly novel and critical fairness issue.

RQ4: Across DECODINGTRUST dimensions, all models showed very low toxicity and perfect ethical compliance, strong privacy protection with zero violations, but varying fairness performance due to demographic and positional biases. GPT-2 lacked meaningful evaluation due to format failure.

RQ5: Model size did not directly correlate with performance; despite being smaller, Mistral-7B outperformed Qwen3-8B due to superior instruction tuning. GPT-2 was unable to follow structured prompts, confirming that architecture and training strategy are more important than parameter count alone.

RQ6: All models consistently demonstrated ethical reasoning, correctly rejecting phishing-related prompts, indicating strong baseline moral alignment across architectures.

RQ7: Model choice had a major impact on data quality, bias behavior, and analytical validity. Instruction-tuned models are essential for cybersecurity research, while base models are unsuitable for structured vulnerability assessment tasks.

6.2 Interpretation of Findings

The education paradox suggests that higher education does not ensure phishing resistance and may increase vulnerability due to overconfidence and targeted attacks. Gender disparity points to the amplification of societal stereotypes within LLM reasoning. The IT domain paradox reflects realistic threat modeling where high-access users are more likely targets, while positional bias indicates a flaw where models favor earlier inputs over true comparative reasoning. The virtual world context enhances realism by enabling contextual reasoning across social and environmental settings.

6.3 Comparison with Prior Work

Findings align with Vishwanath et al. (2018) on personality-driven susceptibility but contradict Sheng et al. (2010) regarding education effects, suggesting a shift toward more sophisticated spear-phishing targeting educated users. Gender bias findings are consistent with Abid et al. (2021), though severity is moderate due to instruction tuning. Positional bias represents a novel contribution not widely reported in prior phishing literature.

6.4 Limitations and Future Work

This study is constrained by limited computation (Colab Pro, 100 units), restricting the dataset to 90 samples and reducing statistical power, with GPT-2 serving only as a baseline. Evaluation relies on model outputs without human validation, includes partial factuality checks, and lacks adversarial robustness testing. Parsing issues caused some data loss (Qwen3-8B), and results may vary with different prompts or model selections.

Future work should expand the dataset, include newer models (e.g., Llama-3, Gemma-2), and introduce adversarial testing (prompt injection, jailbreaks).

7 References

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